END-TO-END

Sentiment Analysis

on Customer Reviews

Advertising & Digital Marketing Industry

CLIENT	DATASET	Prepared By
Candy Factory Group – Pannipitiya	Trustpilot Reviews (June 2022)	Ruwan Pathiranage Sanduni Nisansala

Data Science Internship Project

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Project Overview

End-to-end NLP for Candy Factory Group – Pannipitiya

What & Why

Candy Factory Group – Pannipitiya
A Sri Lankan advertising & digital marketing company.

Goal: Understand customer sentiment in the ad & marketing industry using public Trustpilot reviews, identify themes in positive and negative feedback, and deliver actionable business recommendations.

Deliverables: NLP pipeline · Model comparison report · Sentiment insights · Business recommendations · Presentation

Collect public review dataset

NLP preprocessing pipeline

Exploratory text analysis

Train & compare ML + Transformer

Dataset

Trustpilot Reviews Dataset — June 2022 (Kaggle)

3,698	3,585	214	11
Raw Reviews	After Cleaning	Companies	Columns

Sentiment Label Mapping

Rating	Sentiment	Count	Share
4–5 Stars	**Positive**	2,873	80.1%
3 Stars	**Neutral**	64	1.8%
1–2 Stars	**Negative**	648	18.1%

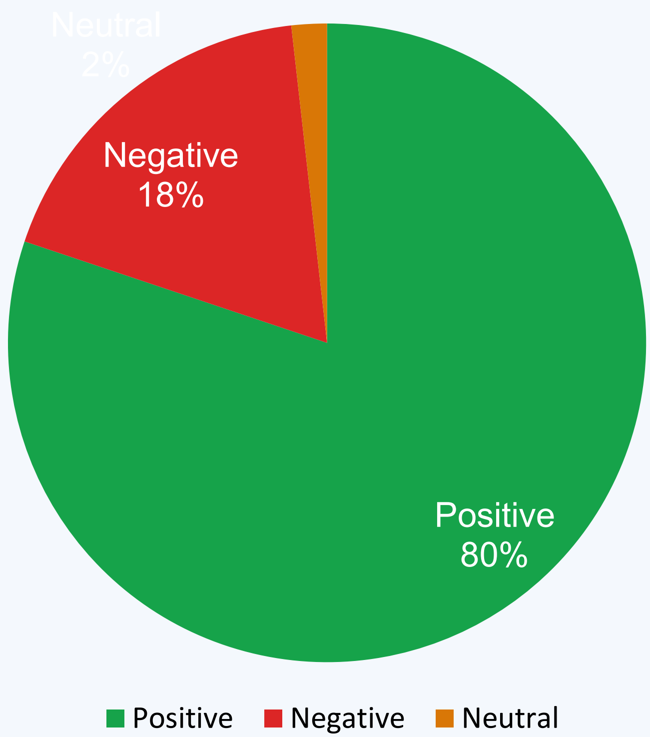
Methodology NLP Pipeline

End-to-end flow from raw data to business insights

1. **Data Collection**
   Kaggle CSV
   3,698 reviews
2. **Data Cleaning**
   Nulls · Dupes
   Outliers removed
3. **Text Pre processing**
   Clean · Tokenize
   SW · Lemmatize
4. **EDA**
   Word clouds
   Distributions
5. **ML Models**
   LR · NB · SVM
   TF-IDF features
6. **Transformer**
   DistilBERT
   Fine-tuning
7. **Insights & Recs**
   Themes · Actions
   Report

Key Findings - Sentiment Distribution

Strong positive skew — but negative reviews are 2.3× longer



80.1% **Positive**
2,873 reviews
Customers praise service quality, speed & professionalism

18.1% **Negative**
648 reviews
Complaints about non-delivery, refunds & poor communication

1.8% **Neutral**
64 reviews
Hardest class - linguistically ambiguous, very underrepresented

Theme Analysis — Word Clouds

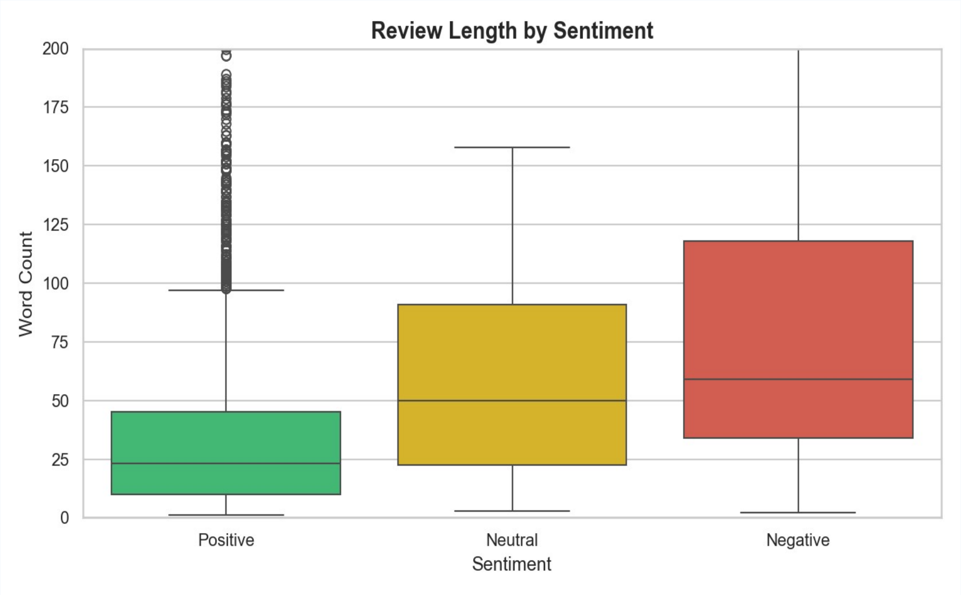
Most frequent words in positive vs negative reviews (stop-words removed)

Positive themes: service · excellent · recommend · helpful · quality · fast · professional

Negative themes: refund · money · never · email · response · bad · delay · charge

Review Length Insight

Dissatisfied customers write more - 2.3× longer than positive reviews



80.2 words
Avg Negative Review Length

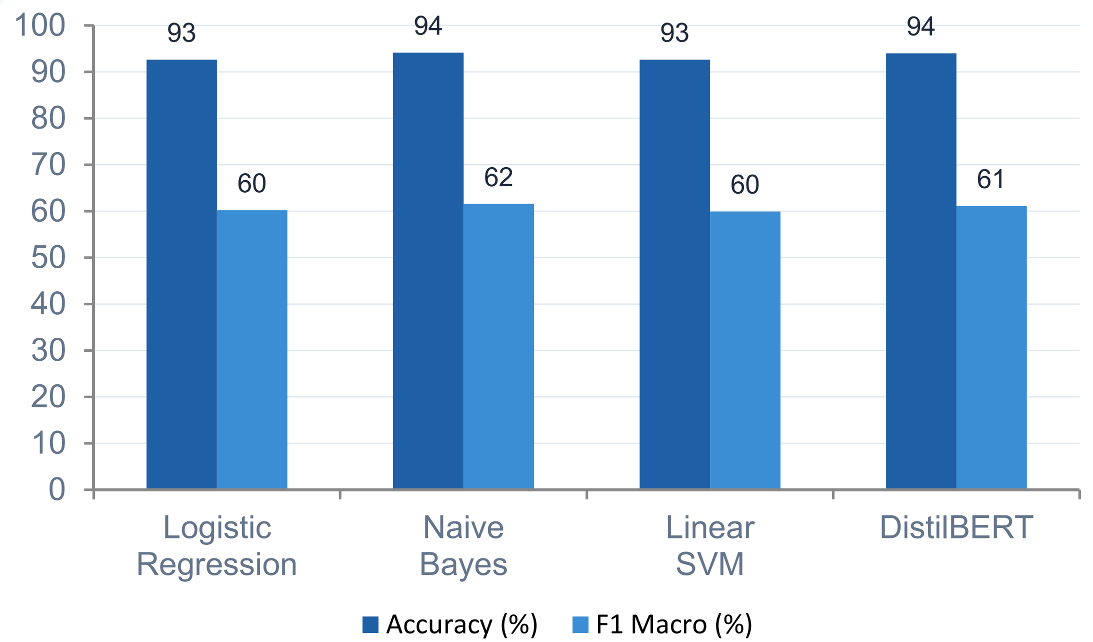
63.7 words
Avg Neutral Review Length

35.4 words
Avg Positive Review Length

Key insight: *Negative reviews contain detailed complaints, making them rich with actionable feedback. Positive reviews are brief endorsements - customers are satisfied and move on.*

Model Performance Results

All models evaluated on same held-out test set (20%, stratified, seed=42)



Model	Accuracy	F1 Macro	CV F1 Mean
Logistic Regression	92.61%	0.6021	0.6035
Naive Bayes	**94.14%**	**0.6153**	**0.6057**
Linear SVM	92.61%	0.5994	0.5993
DistilBERT	94.00%	0.6110	N/A (CPU)

Best model: Naive Bayes (94.14% accuracy, 0.6153 F1 Macro) - fastest training, lightest footprint

ML vs Transformer — Trade-off Analysis

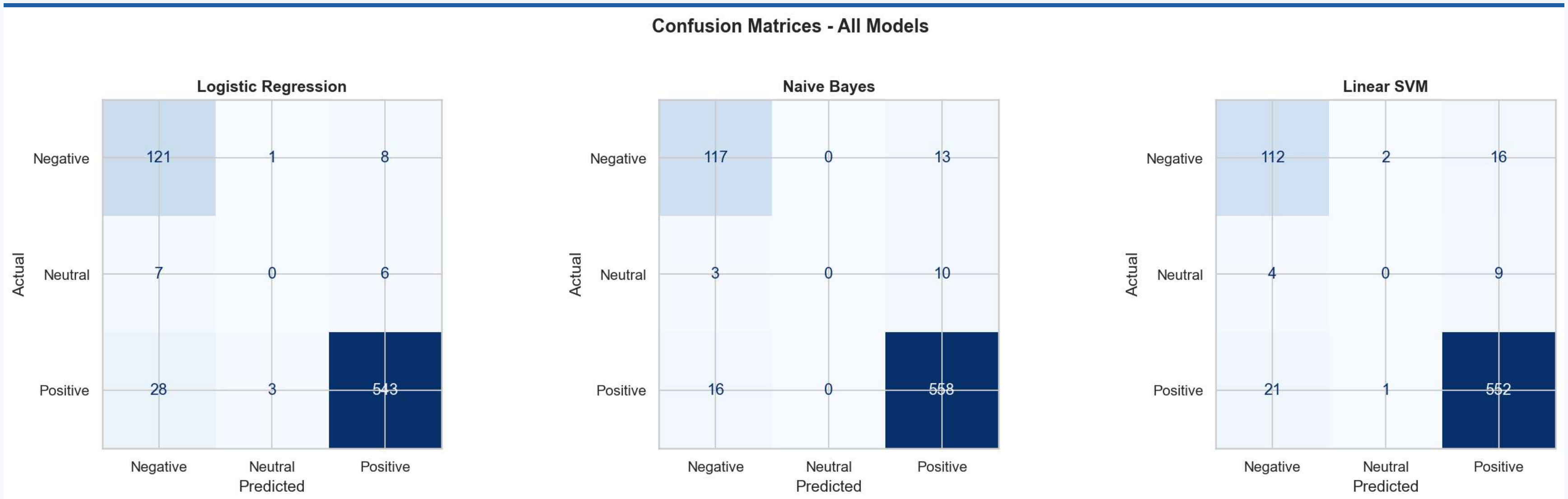
Naive Bayes wins on CPU; DistilBERT recommended for GPU environments

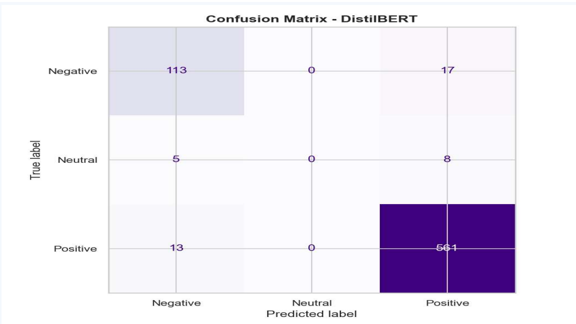
	Logistic Regression	Naive Bayes	Linear SVM	DistilBERT
Training Time	< 5 sec	< 2 sec	< 10 sec	~35 min
Interpretability	★★★★★	★★★★	★★★	★★
Best For	Baseline & feature analysis	Production CPU / real-time	High-dim text, no GPU	Best accuracy with GPU

Recommendation: Deploy Naive Bayes + TF-IDF now (CPU). Retrain DistilBERT with GPU for 5+ epochs when cloud access is available.

Confusion Matrices

Where models make mistakes — Neutral class is hardest to classify





Key Challenges Faced

Three core difficulties impacted modelling performance

Severe Class Imbalance

80.1% of reviews are Positive. A naive model predicting Positive for everything achieves 80% accuracy. This is why macro F1 — not accuracy — is the real metric. Addressed with class_weight='balanced' in LR and SVM.

Tiny Neutral Class

Only 64 Neutral reviews (1.8%). All models struggle to correctly classify this class. It's linguistically ambiguous — 3-star reviews often contain both positive and negative language. Fix: collect more 3-star data or use augmentation.

CPU Constraint — DistilBERT

Windows CPU environment required heavy layer freezing (only Layer 5 unfrozen). Result: DistilBERT F1 flat at 0.611 across all 3 epochs. A Windows DataLoader bug (num_workers crash) also required num_workers=0. Fix: retrain on GPU.

What Customers Love

Themes from 2,873 positive reviews — leverage these in marketing

Service Quality

excellent · great · helpful · professional

Clients value responsive, knowledgeable service delivery above all else.

Team & People

team · friendly · staff · communication

Human interaction quality strongly influences positive sentiment.

Speed & Ease

fast · quick · easy · smooth

Timely delivery and a frictionless client experience drive satisfaction.

Loyalty Intent

again · return · definitely · recommend

High satisfaction converts to expressed loyalty and word-of-mouth.

Tangible Results

recommend · quality · effective · value

Clients praise measurable outcomes - ROI, reach, and conversions.

What Customers Complain About

Themes from 648 negative reviews — these are the pain points to fix

No Response / Ignored

email · never · response · contact · ignored

Clients contact the company and receive no acknowledgement — top complaint.

Communication Failure

call · reply · message · wait · ignored

Breakdown in client communication during or after project delivery.

Refund & Billing Issues

money · refund · charge · pay · dispute

Unexpected charges and difficulty obtaining refunds generate anger.

Delivery Delays

day · week · month · late · still · never

Unmet deadlines and slow delivery are major dissatisfaction drivers.

Poor Service Quality

bad · terrible · waste · disappointment

Service outcomes failed to meet advertised expectations.

Business Recommendations

7 strategies to improve customer satisfaction at Candy Factory Group

24-hr Response SLA
Now

Guarantee email response within 24 hrs. Set auto-acknowledgements.

Post-Project Survey
1–3 mo

Send 3-day post-project satisfaction survey. Track NPS monthly.

Transparent Billing Policy
Now

Publish refund policy. Itemised invoices before charging.

GPU DistilBERT Retrain
3–12 mo

Retrain all layers on Google Colab GPU. Target F1 > 0.75.

Deploy Sentiment Monitor
Now

Run Naive Bayes model on new reviews. Alert team on negatives.

Balance Neutral Class
3–12 mo

Collect 3-star feedback. Augment Neutral data. Retrain models.

Leverage Positive Language
1–3 mo

Use top positive words in marketing copy and testimonials.

Implementation Roadmap

Phased approach to deploying recommendations

Immediate (0–30 days)	Short-Term (1–3 months)	Long-Term (3–12 months)
1 Set 24-hr email response SLA	1 Use positive themes in marketing copy	1 GPU retrain DistilBERT (target F1>0.75)
2 Publish refund & billing policy	2 Launch post-project NPS survey	2 Balance Neutral class via augmentation
3 Deploy Naive Bayes sentiment monitor	3 Track weekly sentiment scores	3 Retrain models every 6 months

Conclusion

What we accomplished and what it means for Candy Factory Group

3,585	4	94.1%	7
Reviews Analysed	Models Trained	Best Accuracy	Business Recs

Complete NLP pipeline built.

Naive Bayes (94.1% accuracy, F1=0.615) is the recommended production model for CPU environments.

Positive reviews cluster around service quality and speed. Negative reviews centre on communication failures and billing disputes.

DistilBERT achieved comparable accuracy (94%) but is limited by CPU. GPU retraining expected to push F1 above 0.75.

Thank You

Sentiment Analysis on Trustpilot Reviews

Candy Factory Group – Pannipitiya | Data Science Internship | June 2026

Questions & Discussion